

Chemicals

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Dedly Chemicals

The first thing we are going to do is make a dataset containing all rows dealing with the 6 deadly chemicals.

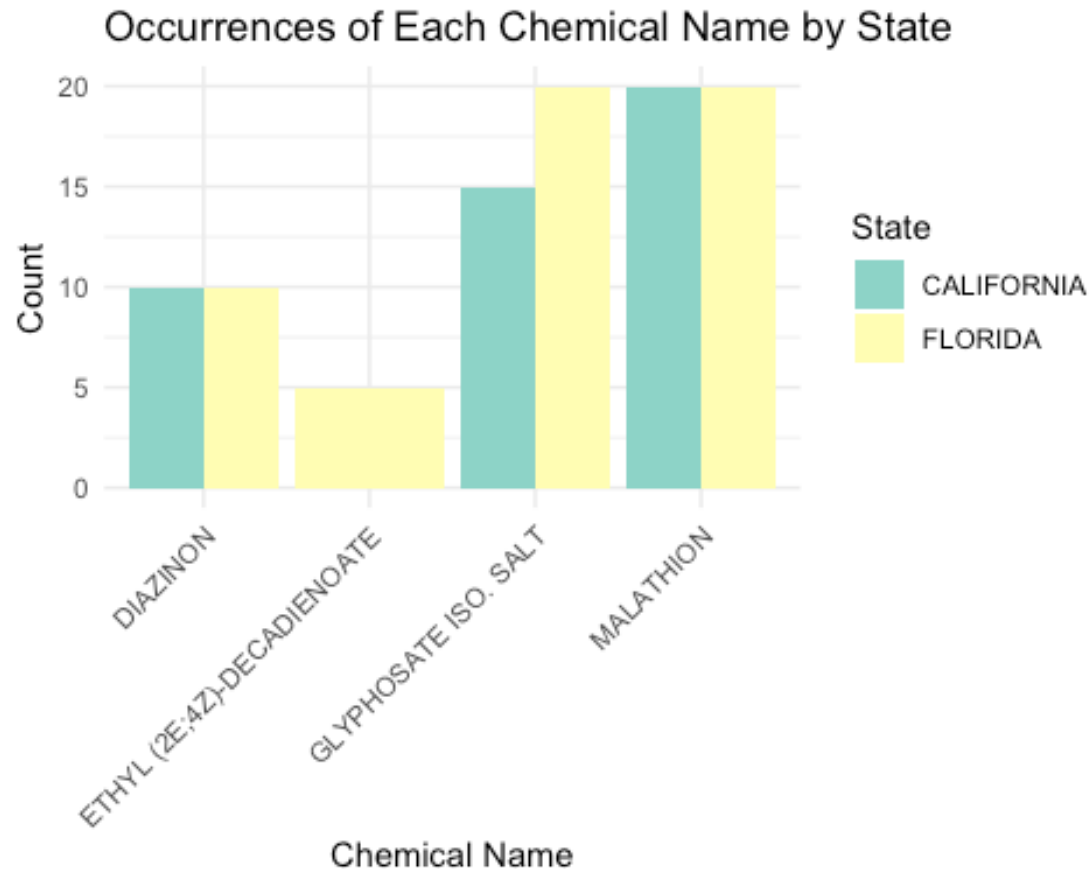
```
sur_chem = read.csv("sur_chem.csv")
deadly_chemicals <- c("CAPTAFOL", "ETHYL (2E;4Z)-DECADIENOATE", "GLYPHOSATE ISO. SALT", "MALATHION", "DIAZINON")
deadly <- subset(sur_chem, chemical_name %in% deadly_chemicals)
```

I then want to see how many times these chemicals come up and break it down by state

```
chemical_counts <- deadly %>%
  group_by(State, chemical_name) %>%
  summarise(count = n(), .groups = 'drop')
```

This bar chart shows us how many times each chemical name shows up in our new dataframe. We can see that the only states that use any of these chemicals are California and Florida which makes sense as that is where the most strawberries are grown

```
ggplot(chemical_counts, aes(x = chemical_name, y = count, fill = State)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Occurrences of Each Chemical Name by State", x = "Chemical Name", y = "Count") +
  theme_minimal() +
  scale_fill_brewer(palette = "Set3") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Malathion

For this EDA i am focusing on the chemical Malathion as it contains the most data of the chemicals listed here. What i found interesting is that there is a value indicting the percent of area bearing strawberries in California that use this chemical as an Insecticide by year. We are going to use the most recent year from this (2023)

We are going to use the California average of 40,000 acres of strawberries grown per year for some quick math

```
mal_cal_pct = 40000 * .19
print(mal_cal_pct)

## [1] 7600
```

So 7600 acres of strawberries in california used this chemical as an insecticide. So what does that mean?

Well let's compare it to all insecticide's used in california, we can make a dataframe for that

```
california_insecticide <- subset(sur_chem, State == "CALIFORNIA" & use == "INSECTICIDE")
```

use unique to figure out how many insecticides were used in califoria

```
unique(california_insecticide$chemical_name)
```

```
## [1] "CYCLANILIPROLE"
## [2] "PERMETHRIN"
## [3] "ABAMECTIN"
## [4] "ACEQUINOCYL"
## [5] "ACETAMIPRID"
## [6] "AZADIRACHTIN"
## [7] "BEAUVERIA BASSIANA"
## [8] "BIFENAZATE"
## [9] "BIFENTHRIN"
## [10] "BT KURSTAK ABTS-1857"
## [11] "BT KURSTAKI ABTS-351"
## [12] "BT KURSTAKI SA-11"
## [13] "CANOLA OIL"
## [14] "CHLORANTRANILIPROLE"
## [15] "CHROMOBAC SUBTSUGAE PRAA4-1 CELLS AND SPENT MEDIA"
## [16] "CYANTRANILIPROLE"
## [17] "CYFLUMETOFEN"
## [18] "ETOXAZOLE"
## [19] "FENBUTATIN-OXIDE"
## [20] "FENPROPATHRIN"
## [21] "FENPYROXIMATE"
## [22] "FLONICAMID"
## [23] "FLUPYRADIFURONE"
## [24] "HEXYTHIAZOX"
## [25] "IMIDACLOPRID"
## [26] "LAMBDA-CYHALOTHRIN"
## [27] "MALATHION"
## [28] "METHOXYFENOZIDE"
## [29] "NALED"
## [30] "NEEM OIL"
## [31] "NOVALURON"
## [32] "PIPERONYL BUTOXIDE"
## [33] "PYRETHRINS"
## [34] "PYRIDABEN"
## [35] "SPINETORAM"
## [36] "SPINOSAD"
## [37] "THIAMETHOXAM"
## [38] "CHEMICAL"
## [39] "BT KURSTAKI EG7841"
## [40] "BT SUB AIZAWAI GC-91"
## [41] "BUPROFEZIN"
## [42] "BURKHOLDERIA A396 CELLS & MEDIA"
## [43] "DIAZINON"
```

```
## [44] "HELICOVERPA ZEA NPV"
## [45] "PETROLEUM DISTILLATE"
## [46] "POTASSIUM SALTS"
## [47] "PYRIPROXYFEN"
## [48] "SPIROMESIFEN"
## [49] "CARBARYL"
## [50] "CHLORPYRIFOS"
## [51] "SOYBEAN OIL"
## [52] "SULFOXAFLO"
## [53] "ZETA-CYPERMETHRIN"
## [54] "EMAMECTIN BENZOATE"
## [55] "SPIROTETRAMAT"
```

This tells us there are 55 different insecticides used. That seems like too many to compare so let's get the top 5

```
# Replace (D) with NA in the 'Value' column of california_insecticide
california_insecticide$Value[california_insecticide$Value == "(D)"] <- NA

# Remove commas and convert Value to numeric
california_insecticide$Value <- as.numeric(gsub(",", "",
california_insecticide$Value))

## Warning: NAs introduced by coercion

# Remove rows where Value is NA
california_insecticide <- california_insecticide %>% filter(!is.na(Value))

california_insecticide$category <- trimws(california_insecticide$category)

top_chemicals <- california_insecticide %>%
  filter(category == "BEARING - TREATED",
    measurement == "PCT OF AREA BEARING, AVG",
    chemical_name != "CHEMICAL",
    Year == 2023) %>% # Include only rows where year is 2023
  arrange(desc(Value)) %>% # Arrange by Value in descending order
  select(chemical_name, Value) %>% # Select only chemical_name and Value
columns
head(5)

print(top_chemicals)
```

##	chemical_name	Value
## 1	FLONICAMID	73
## 2	SPINETORAM	70
## 3	METHOXYFENOZIDE	66
## 4	BIFENAZATE	61
## 5	BIFENTHRIN	60

so this gives us our top 5 insecticides in California based on percent used of bearing treated acres of strawberries.

Effects of These Insecticides

We know that Malathion is one of the 6 deadly chemicals that is used in strawberries, but what are its actual effects and how does it compare to these other insecticides?

We have built a function that used PubChem to look up the effects of these chemicals. I displayed the results for Malathion below.

```
library(tidyverse)

## — Attaching core tidyverse packages ————— tidyverse
## 2.0.0 —
## ✓ forcats 1.0.0      ✓ stringr 1.5.1
## ✓ lubridate 1.9.3    ✓ tibble 3.2.1
## ✓ purrr 1.0.2       ✓ tidyr 1.3.1
## ✓ readr 2.1.5
## — Conflicts —————
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag() masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors

library(PubChemR)

GHS_searcher<-function(result_json_object){
  result<-result_json_object
  for (i in 1:length(result[["result"]][["Hierarchies"]][["Hierarchy"]])){
    if(result[["result"]][["Hierarchies"]][["Hierarchy"]][i][["SourceName"]]==
      "GHS Classification (UNECE)"){
      return(i)
    }
  }
}

hazards_retriever<-function(index,result_json_object){
  result<-result_json_object
  hierarchy<-result[["result"]][["Hierarchies"]][["Hierarchy"]][index]
  i<-1
  output_list<-rep(NA,length(hierarchy[["Node"]]))
  while(str_detect(hierarchy[["Node"]][i][["Information"]][["Name"]], "H") &
    i<length(hierarchy[["Node"]])){
    output_list[i]<-hierarchy[["Node"]][i][["Information"]][["Name"]]
```

```

    i<-i+1
  }
  return(output_list[!is.na(output_list)])
}

result_d<-get_pug_rest(identifier = "Malathion", namespace = "name", domain =
"compound",operation="classification", output = "JSON")

hazards_retriever(GHS_searcher(result_d),result_d)

## [1] "H302: Harmful if swallowed [Warning Acute toxicity, oral]"
## [2] "H300: Health Hazards"
## [3] "Hazard Statement Codes"
## [4] "H317: May cause an allergic skin reaction [Warning Sensitization,
Skin]"
## [5] "H320: Causes eye irritation [Warning Serious eye damage/eye
irritation]"
## [6] "H331: Toxic if inhaled [Danger Acute toxicity, inhalation]"
## [7] "H341: Suspected of causing genetic defects [Warning Germ cell
mutagenicity]"
## [8] "H350: May cause cancer [Danger Carcinogenicity]"
## [9] "H370: Causes damage to organs [Danger Specific target organ
toxicity, single exposure]"
## [10] "H372: Causes damage to organs through prolonged or repeated exposure
[Danger Specific target organ toxicity, repeated exposure]"
## [11] "H373: May causes damage to organs through prolonged or repeated
exposure [Warning Specific target organ toxicity, repeated exposure]"
## [12] "H400: Very toxic to aquatic life [Warning Hazardous to the aquatic
environment, acute hazard]"
## [13] "H400: Environmental Hazards"
## [14] "H410: Very toxic to aquatic life with long lasting effects [Warning
Hazardous to the aquatic environment, long-term hazard]"

```

These are some fairly serious hazards, but what i am going to focus one if one of the effects, which I found was also prevelant in the insectisides Methoxyfenozide and Bidenazate from our list as well, which is that they are very toxic to aquatic life.

Aquatic Chemicals EDA

Let's break down some more numbers

After some googling, the average number of pounds of strawberries produced per acre is about 50,000 pounds. So let's compare some numbers. first let's add Malathion to our data_frame and create a new table called aquatic_chemicals with the three we are looking at.

```

new_row <- data.frame(chemical_name = "MALATHION", Value = 19)

# Add the new row to the top_chemicals dataframe

```

```

top_chemicals <- rbind(top_chemicals, new_row)

print(top_chemicals)

##      chemical_name Value
## 1      FLONICAMID    73
## 2      SPINETORAM    70
## 3 METHOXYFENOZIDE    66
## 4      BIFENAZATE    61
## 5      BIFENTHRIN    60
## 6      MALATHION     19

aquatic_chemicals <- top_chemicals %>%
  filter(chemical_name %in% c("MALATHION", "METHOXYFENOZIDE", "BIFENAZATE"))

```

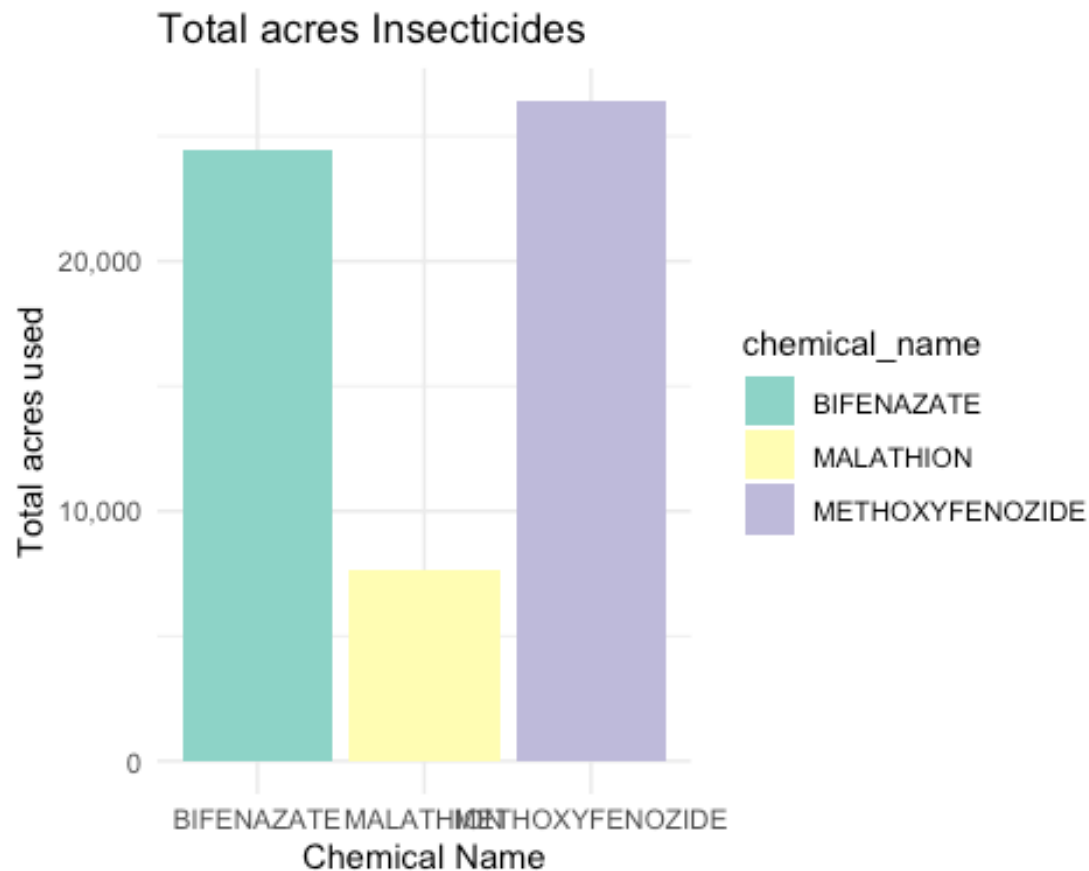
Let's first make a chart showing the total number of acres that used these chemicals in 2023

```

aquatic_chemicals$total_acres <- (aquatic_chemicals$Value / 100) * 40000

ggplot(aquatic_chemicals, aes(x = chemical_name, y = total_acres, fill =
chemical_name)) +
  geom_bar(stat = "identity") +
  labs(title = "Total acres Insecticides",
       x = "Chemical Name",
       y = "Total acres used") +
  theme_minimal() +
  scale_y_continuous(labels = scales::comma) + # Format y-axis labels with
commas
  scale_fill_brewer(palette = "Set3")

```



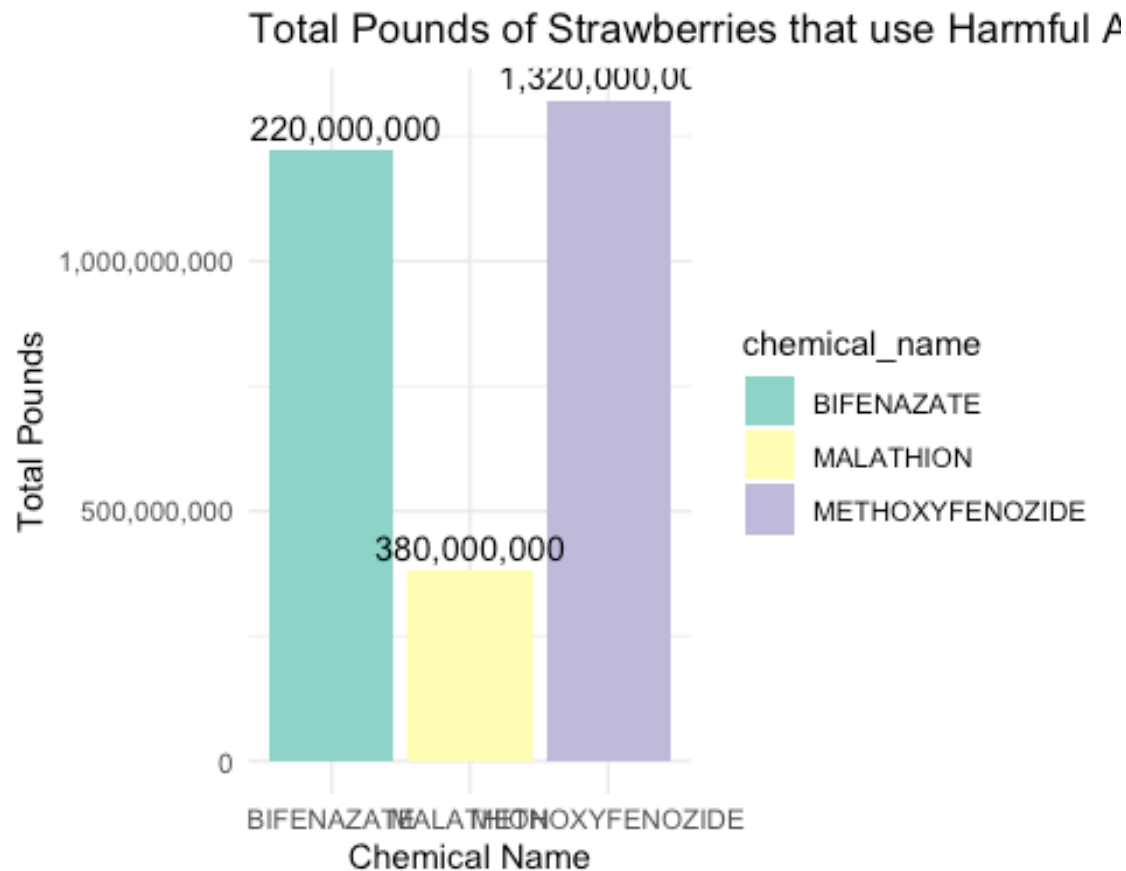
we can add another value to this dataframe showing the total pounds of strawberries using these chemicals. And a chart that displays the total pounds of strawberries that use these chemicals.

```

aquatic_chemicals$total_pounds <- (aquatic_chemicals$total_acres) * 50000

ggplot(aquatic_chemicals, aes(x = chemical_name, y = total_pounds, fill =
chemical_name)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = scales::comma(total_pounds)), vjust = -0.5) + # Add
Labels with commas
  labs(title = "Total Pounds of Strawberries that use Harmful Aquatic
Insecticides",
        x = "Chemical Name",
        y = "Total Pounds") +
  theme_minimal() +
  scale_y_continuous(labels = scales::comma) + # Format y-axis labels with
commas
  scale_fill_brewer(palette = "Set3") # Optional: color palette for the bars

```



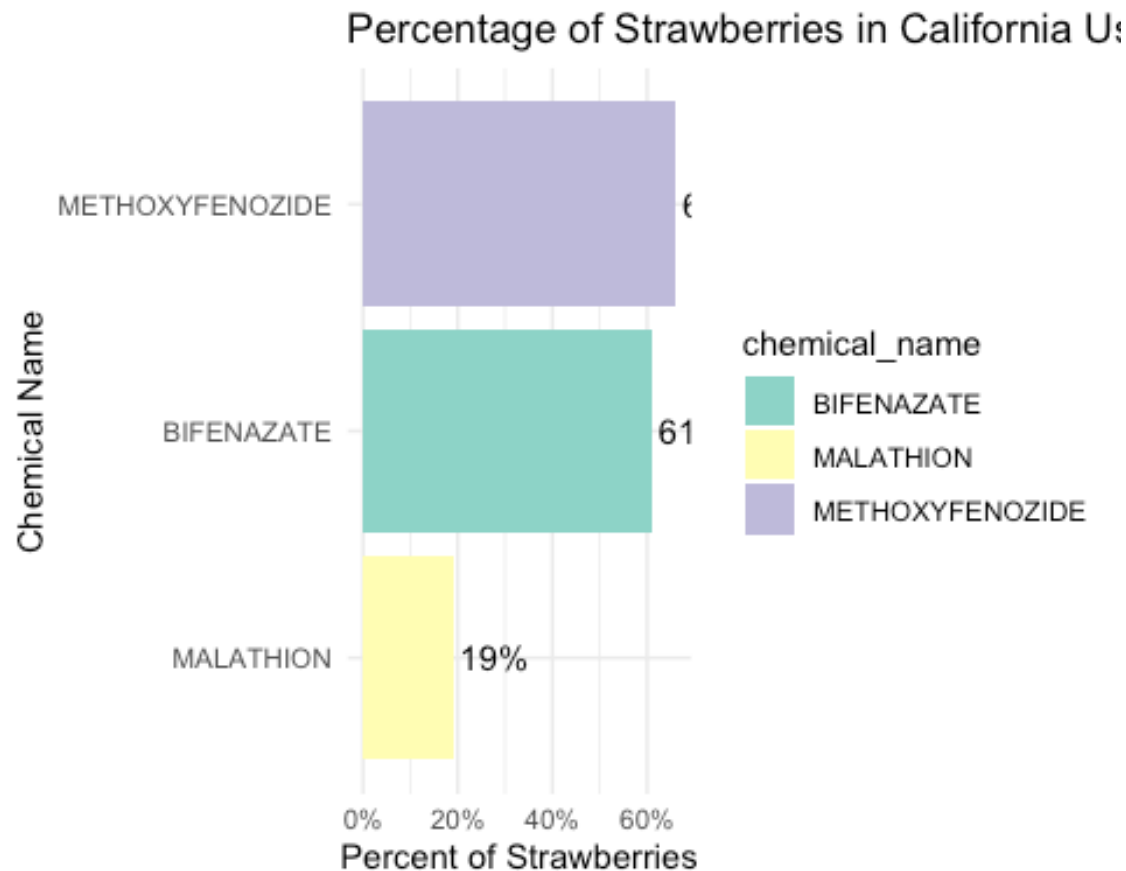
And the percentages of strawberries grown that use these chemicals

```

aquatic_chemicals$total_percent <- (aquatic_chemicals$total_pounds /
2000000000) * 100

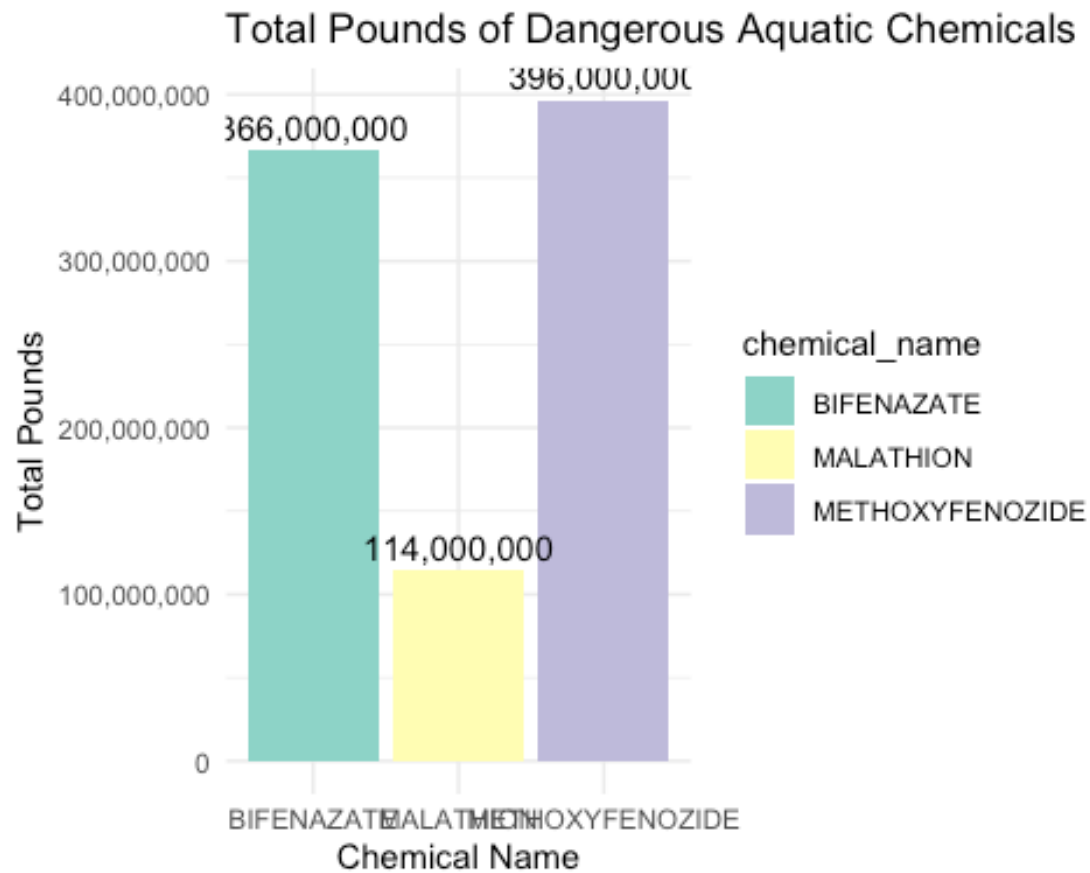
ggplot(aquatic_chemicals, aes(x = reorder(chemical_name, total_percent), y =
total_percent, fill = chemical_name)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = paste0(round(total_percent, 0), "%")), hjust = -0.1)
+ # Add percentage labels next to the bars
  labs(title = "Percentage of Strawberries in California Using These
Insecticides",
        x = "Chemical Name",
        y = "Percent of Strawberries") +
  theme_minimal() +
  scale_y_continuous(labels = scales::percent_format(scale = 1)) + # Format
y-axis labels as percentages
  scale_fill_brewer(palette = "Set3") +
  coord_flip()

```



However this does not tell the full story. According to research, there are about .3 pounds of chemicals used for every pound of strawberries grown. So let's update the total pounds chart.

```
ggplot(aquatic_chemicals, aes(x = chemical_name, y = total_pounds * 0.3, fill = chemical_name)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = scales::comma(round(total_pounds * 0.3, 0))), vjust = -0.5) + # Add labels above bars
  labs(title = "Total Pounds of Dangerous Aquatic Chemicals",
        x = "Chemical Name",
        y = "Total Pounds") +
  theme_minimal() +
  scale_y_continuous(labels = scales::comma) + # Format y-axis labels with commas
  scale_fill_brewer(palette = "Set3") # Optional: color palette for the bars
```

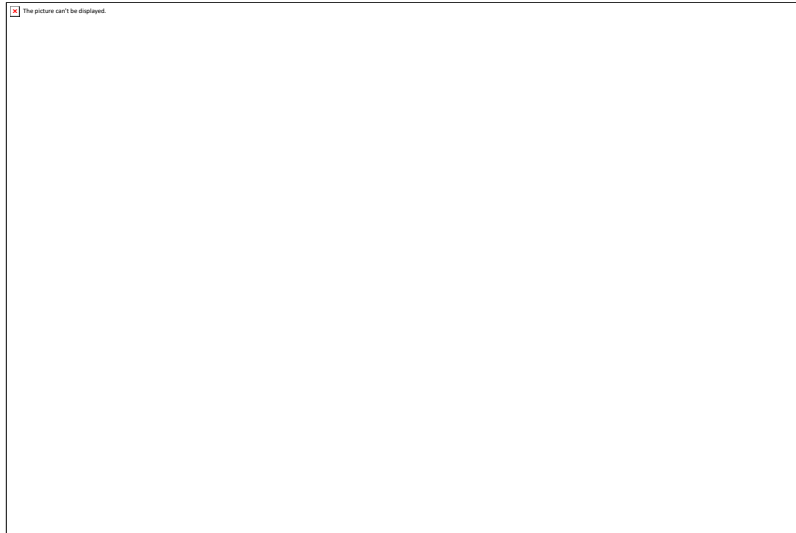


So added together that is 876 Million Pounds of these harmful aquatic chemicals used on Strawberries in California in 2023.

Consequences

So what is the effect of 876 million pounds of these chemicals?

Since A large amount of California's strawberries are grown on the coast via the image below



California Strawberry Growth.

This could potentially be very dangerous to the Aquatic life off the coast of California. However the main way that these chemicals could get into the ocean would be via rain. Which is very low in the area.

In fact the average rainfall in California is between 2018 and 2023 was just over 21 inches per year as seen in this chart.

```
rainfall <- read_csv("data-BEa5S.csv")

## Rows: 128 Columns: 2
## — Column specification
## Delimiter: ","
## dbl (2): Year, Precipitation (inches)
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

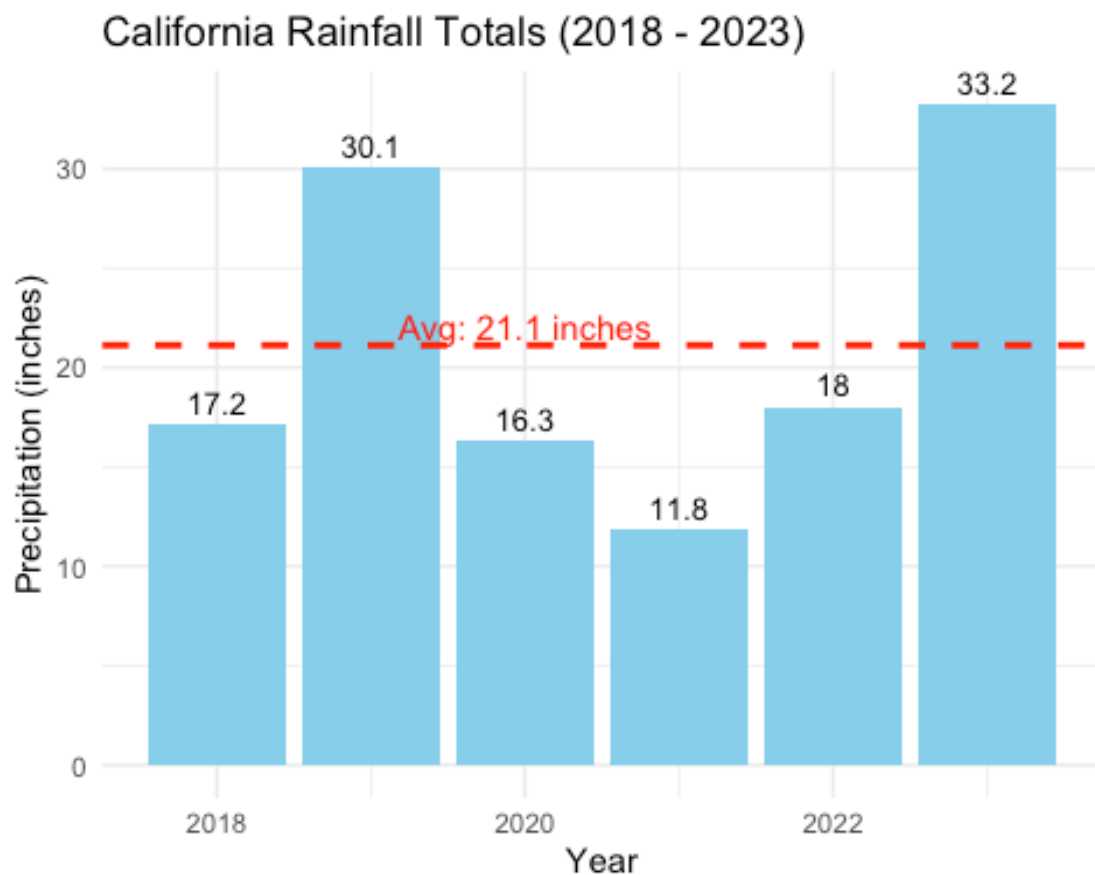
rainfall_filtered <- rainfall %>%
  filter(Year >= 2018 & Year <= 2023)

# Calculate the average precipitation for these years
average_precip <- mean(rainfall_filtered$`Precipitation (inches)`, na.rm = TRUE)

# Create the chart
ggplot(rainfall_filtered, aes(x = Year, y = `Precipitation (inches)`) +
  geom_bar(stat = "identity", fill = "skyblue") +
  geom_hline(yintercept = average_precip, color = "red", linetype = "dashed",
    size = 1) +
  geom_text(aes(label = round(`Precipitation (inches)`, 1)), vjust = -0.5,
```

```
size = 3.5) + # Show values on top of bars
  labs(title = "California Rainfall Totals (2018 - 2023)",
        x = "Year",
        y = "Precipitation (inches)") +
  annotate("text", x = 2020, y = average_precip + 1, label = paste("Avg:",
round(average_precip, 1), "inches"), color = "red") +
  theme_minimal()

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



That is a rate of just .05 inches of rainfall per day. According to studies it takes over an inch of rainfall to wash away any pesticides off of plants and fruits meaning that these dangerous insecticides put on Strawberries in California are most likely not washed into nearby rivers, lakes or oceans.